

HealthConnect Week 6 Project Summary

Data Analytics Track

To: HealthConnect Experience Lab

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❖ What I planned to accomplish

Deepen the strongest Week 5 findings, validate important KPIs, investigate interactions, integrate Data Science outputs, improve the dashboard and prepare analytical testing requirements for Week 7.

❖ What I completed

I revalidated the eligible appointment population and 51.15% no-show baseline, refined lead-time bands, performed advanced segment and interaction analysis, added statistical validation, reviewed Data Science model outputs, recalculated metrics from the supplied scoring file and designed a two-page Week 6 dashboard update.

❖ What improved from Week 5

Week 5 primarily described one-variable patterns. Week 6 tested whether important factors remained meaningful, quantified association strength and investigated combinations such as lead time × previous no-show history.

❖ What I integrated

Data Science feature/model outputs were incorporated into the analytical workflow. Analytics findings were also shared back to Data Science to support feature refinement and error analysis.

❖ Track(s) collaborated with

Data Science.

❖ What was exchanged

Received: Random Forest feature/threshold/error information, Logistic Regression coefficients and a scoring file. Provided: validated findings on lead time, previous no-show history, reminders, distance, appointment type and priority segments.

❖ What changed as a result

The analysis shifted from independent EDA to model-informed decision support. Lead time and previous no-show history became the primary cross-track focus, while weaker variables such as age, day and time were deprioritised from the main intervention story.

❖ **Key findings/outcomes**

- Overall no-show rate remained 51.15%.
- No-show rate increased from 29.5% at 0–7 days to 68.5% at 41–60 days.
- 41–60 day lead time combined with 2+ prior no-shows recorded 83.3% no-shows.
- 20+ km distance recorded 60.2% no-shows.
- No reminder recorded 54.6% vs 49.9% with reminder.
- The Logistic Regression scoring output achieved approximately 68.9% ROC-AUC.

❖ **Major challenges**

The scoring file does not preserve the true HealthConnect appointment ID, some engineered feature definitions remain unverified, and outputs from different Data Scientists cannot be directly compared without confirming the same evaluation design.

❖ **Important decisions**

I retained the Week 5 no-show denominator, aligned lead-time bands with Data Science, treated model thresholds as decision trade-offs rather than fixed business rules, and avoided joining the scoring file to raw data using the integer row ID.

❖ **Remaining issues**

True appointment IDs are required in the next scoring export; operational threshold costs/capacity remain undefined; exact engineered feature definitions should be confirmed from code.

❖ **Contribution to HealthConnect**

The Analytics track now provides validated priority segments and a business interpretation layer that connects descriptive appointment patterns with predictive model outputs.

❖ **Proposed focus for Week 7**

Perform row-level model error analysis by operational segment, test dashboard measures, confirm feature definitions, support threshold selection and validate the stability of the highest-priority segments.